

Business Problem Statement — ShopPulse

Retail Customer Intelligence & Purchase Analytics

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management has observed differences in purchasing patterns across customer demographics, product categories, discounts, reviews, seasons, and payment preferences.

The company initially seeks to analyze its consumer shopping data to identify trends, understand purchasing behavior, and support data-driven marketing and product decisions.

To extend this analysis, management wants to develop a deeper understanding of **customer value and segment performance**. Rather than treating all customers as a single group, the company wants to identify meaningful customer segments based on purchasing behavior and evaluate how these segments differ in commercial value.

Customer Intelligence Questions

1. **Customer Value Segmentation**
How can customers be segmented based on their **purchase value and previous purchase history**?
2. **Customer Base Composition**
What proportion of the customer base belongs to each identified customer value segment?
3. **Segment Spending Behavior**
How does **average purchase amount** differ across the identified customer segments?
4. **Segment Performance**
How do the customer segments compare in terms of **customer volume, average purchase amount, customer ratings, and revenue contribution**?

Customer Segments

The analysis categorizes customers into four value-oriented segments:

- **VIP** — high purchase value and high purchase history
- **Growth** — high purchase value but lower purchase history
- **Loyalist** — lower purchase value but high purchase history
- **Low Engagement** — lower purchase value and lower purchase history

Business Objective

The objective is to provide management with a clearer view of **who the company's customers are, how valuable different customer groups are, and where customer engagement and business value differ across segments**.

This complements the original Python, SQL, and Power BI analysis and adds a dedicated **Customer Intelligence** layer to the project. The original project already defines Python data preparation, SQL analysis, Power BI visualization, reporting, and GitHub documentation as its core deliverables.